



VANSH AGRAWAL

va2583@srmist.edu.in | +91 8810223022 | 2027

B.Tech, Computer Science and Engineering — AI & ML

Address: SRM IST, Kattankulathur

GitHub: github.com/vansh070605

LinkedIn: linkedin.com/in/thevanshagrawal



PROFILE

Final-year Computer Science student and AI/ML Engineer with hands-on experience building and deploying production-grade machine learning systems, specializing in computer vision, multi-agent RAG architectures, and real-time edge intelligence. Currently working as an AI/ML Intern at Tata Steel, developing scalable computer vision solutions for industrial quality control. Built multiple end-to-end applications across autonomous agent frameworks and real-time vision, focusing on deployable, low-latency architectures.

SKILLS

Languages

Python JavaScript TypeScript SQL Java C / C++ HTML5 / CSS3

Frameworks & Libraries

TensorFlow OpenCV Google MediaPipe SentenceTransformers FastAPI React 19 Scikit-learn Pandas NumPy

Tools & Databases

Git / GitHub Docker ChromaDB (Vector DB) Firebase PostgreSQL MySQL Vercel REST APIs

EXPERIENCE

Tata Steel

Nov 2025 – Present

AI / ML Intern

Chennai, India

- Engineered a Computer Vision pipeline using OpenCV for automated TMT rebar quality control, calculating rib parameters with 95% precision to evaluate strict pass/fail tolerances.
- Developed an automated validation module for chemical ring tests, utilizing spatial analysis to accurately measure rod thickness and reduce manual inspection time by 40%.

RCOTBA Industries

Jan 2025 – Jun 2025

Team Lead

Chennai, India

- Directed cross-functional teams and achieved 100% of agile delivery deadlines.
- Developed custom client websites using React, streamlining internal pipelines to deploy new sites from concept to live hosting in just 2-3 days and increasing delivery speed by 35%.

PROJECTS

PlacementOS — AI Career Command Center

GitHub

React 19 FastAPI ChromaDB SentenceTransformers TensorFlow Firebase

- Developed PlacementOS, a local-first AI career platform powered by 9 specialized autonomous agents. Engineered a local RAG loop using SentenceTransformers and ChromaDB to extract job description metrics and tailor resumes with the Google X-Y-Z formula. Built a career compass, codebase project defense Q&A generator, and a neural network regression model (TensorFlow/Keras) for localized salary band prediction.

WakeGuard — AI Driver Safety Vision System

GitHub

React Vite Google MediaPipe Tailwind CSS Recharts

- Developed WakeGuard, a real-time client-side driver monitoring system tracking 478 3D facial landmarks at 60 FPS. Utilized MediaPipe Face Mesh to calculate geometric EAR (Eye Aspect Ratio), MAR (Mouth Aspect Ratio), and 3D head rotation to detect microsleeps, yawning, or distraction. Implemented voice synthesis warnings via Web Speech API and a glassmorphism telemetry dashboard with zero backend dependencies.

AERIS — Flood Risk Intelligence System

GitHub

XGBoost Python FastAPI React 19 GeoJSON

- Developed AERIS, an AI-powered predictive flood management platform for the Bihar region, utilizing a React/Tailwind frontend and a FastAPI backend to deliver proactive 7-day disaster response intelligence. Integrated dual XGBoost models and a React-Leaflet GIS dashboard to analyze live telemetry across 18 river stations, simulating multi-variate hydraulic stress and automating breach predictions.

EDUCATION

SRM IST, Kattankulathur

B.Tech — CSE (AI & ML)

2023 – 2027 • CGPA: 7.86 / 10

A.P.J School, NOIDA

Class XII — CBSE (MPC)

2023 • 68.4%

A.P.J School, NOIDA

Class X — CBSE

2021 • 85.2%